SARS-CoV infection
ACE2, soluble
S
S
S
HSPA5
HSPA5-Spike interaction
pulmonary fibrosis
S
active
ACE2, membrane-bound
ACE2-SARS-CoV interaction
active
ACE2, membrane-bound
ADAM17
ACE2
viral RNA genome replication
viral entry into host cell
PACS2
ADAM17
PACS2-ADAM17 interaction
early endosome
ACE2:Spike
nucleus
ATF6
MFN2
HSPA5
ACE2
XBP1
BCL2
DDIT3
TRIB3
ATF4
ERO1A
HYOU1
PPP1R15A
DNAJC3
DNAJC3
presence of dsRNA
EIF2AK2
EIF2AK4
ATP
GCN2:ATP
EIF2AK1
EIF2S1
EIF2S2
EIF2S3
GDP
EIF2-P:GDP
EIF2S3:GDP
EIF2:GDP
EIF2:GDP:Met-tRNA
Met-tRNA
GTP
Ternary Complex
EIF2B1
EIF2B2
EIF2B3
EIF2B4
EIF2B5
EIF2B
Translation initiation
DDIT3
DDIT3
XBP1
MAPK14
TRIB3
ATF4
ERO1A
PPP1R15A
DNAJC3
Persistant ER Stress
cleaved~CASP3
CASP3
cleaved~CASP4
CASP4
cleaved~CASP9
CASP9
CAPN1
Ca2+
CAPN1:Ca2+
CAPN1
Ca2+
high Ca2+ cytosolic concentration
release of ER Ca2+
XBP1
ERAD
protein ubiquitination and destruction
ERN1
BCL2L11
ERN1:BCL2L11
ERN1
BBC3
ERN1:BBC3
ERO1A
hyperoxidation
endoplasmic reticulum
TMBIM6
ITPR
RYR1
ERO1A
BAX
ERN1
BAX:ERN1
accumulation of misfolded protein in ER
ER Stress
Persistant ER Stress
retrograde transport from ER to cytosol
Unfolded protein
disulphide bond
MFN2
endoplasmic reticulum organization
Apoptosis
unfolded protein response (UPR)
ER stress
mitochondrion-endoplasmic reticulum membrane tethering
endoplasmic reticulum calcium ion homeostasis
HSPA5
p20
PACS2
autophagy
VAPB
Ca2+ ER depletion
BAX
BAK1
BCL2
BCL2L11
Unfolded protein
ERN1
TRAF2
TRAF2:ERN1
BAK1
ERN1
BAK1:ERN1
EIF2AK3
EIF2AK3:EIF2AK3
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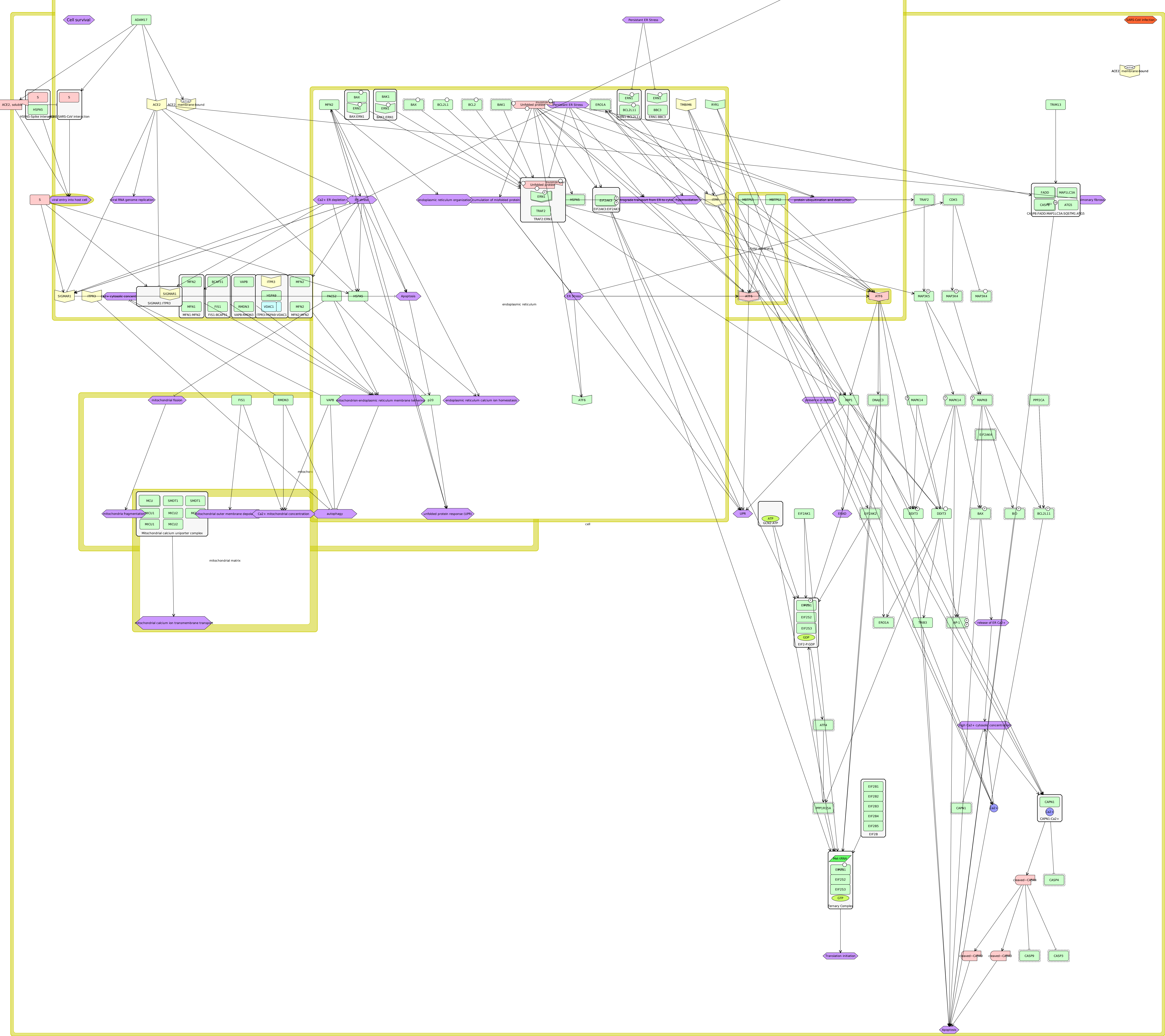
Unfolded protein
MFN2
MFN1
MFN1:MFN2
MFN2
MFN2
MFN2:MFN2
VAPB
RMDN3
VAPB:RMDN3
associated endoplasmic reticulum membrane
Ca2+ cytosolic concentration
BCAP31
FIS1
FIS1:BCAP31
ITPR3
HSPA9
VDAC1
ITPR3:HSPA9:VDAC1
ITPR3
SIGMAR1
SIGMAR1:ITPR3
ITPR3
SIGMAR1
RMDN3
FIS1
mitochondrion
mitochondrial fission
mitochondria fragmentation
mitochondrial outer membrane depolarization
Ca2+ mitochondrial concentration
MCU
SMDT1
SMDT1
MICU1
MICU2
MCUb
MICU1
MICU2
Mitochondrial calcium uniporter complex
mitochondrial matrix
mitochondrial calcium ion transmembrane transport
MCU
SMDT1
MCU:MICU1:MICU2:SMDT1
MCU2
MCU1
TRAF2
MAP3K5
MAPK14
MAPK8
CDK5
MAP3K4
MAP3K4
BAX
BID
BCL2L11
AP-1
PPP2CA
DDIT3
DDIT3
cell
Cell survival
Apoptosis
TRIB3
UPR
XBP1
TRIM13
FADD
MAP1LC3A
CASP8
K63
Ub
ATG5
CASP8:FADD:MAP1LC3A:SQSTM1:ATG5
Golgi apparatus
MBTPS1
MBTPS2
ATF6
ATF6:HSPA5
P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